Dataset Research for Real-Time ASL Translation

Overview

This document covers the datasets we need to train and evaluate the ASL translation device. The goal is real-time translation of ASL signs into English text on the Jetson, so the data has to support video-based sign recognition rather than static image classification.

What We Need in a Dataset

For real-time ASL recognition, the dataset should have:

- **Video-based samples** with labeled ASL signs (static image datasets are not enough since signs depend on motion).
- **Word-level annotations** so each clip maps to a known English word or gloss.
- **Multiple signers** to capture variation in signing styles. Everyone signs a little differently (size of movement, speed, hand shape, left vs right-handed), so training on only one signer limits generalization.
- **Varied lighting and backgrounds** to make the model more robust in real-world use.
- **High frame rate** (24 to 30 fps minimum) so fast motion between sign transitions is captured cleanly.
- **Clean hand visibility** with minimal occlusion, since MediaPipe relies on visible landmarks.

Existing Datasets

WLASL (Word Level American Sign Language)

- 2,000+ signs across 21,000+ videos
- Multiple signers per sign
- Word-level labels
- Widely used as a benchmark for isolated sign recognition
- Good starting point, and the dataset we are currently targeting

MS-ASL (Microsoft ASL)

- 1,000 signs across 25,000+ videos
- Large signer pool (around 200 signers)

- Strong diversity in background and lighting
- Useful as a supplement or for pretraining

How2Sign

- Continuous, sentence-level ASL (not isolated signs)
- Paired with English translations
- Useful later if we move from word-level recognition to full sentence translation
- Probably out of scope for the initial prototype

Additional Steps Beyond the Public Datasets

Public datasets alone may not be enough for a polished demo, so we should plan for:

- **Fine-tuning with custom recordings** of our own hands under the conditions the device will actually be used in (our lighting, camera angle, distance from the Jetson camera, etc.).
- **Data augmentation** to artificially expand the training set. Useful techniques:
    - Horizontal flips (mirror image)
    - Small rotations to simulate different camera angles
    - Speed variation (faster or slower signing)
    - Brightness and contrast shifts for different lighting
- **Targeted data collection** for any signs that are underrepresented or perform poorly during testing.

Initial Recommendation

Start with **WLASL** as the primary dataset since it has the largest vocabulary and is well-documented. Use **MS-ASL** as a secondary source if we need more signer diversity. Hold off on **How2Sign** until the isolated sign pipeline works end-to-end, since sentence-level translation is a much harder problem and not required for the first working demo.

After the base model is trained, record a small custom dataset on the actual Jetson camera and fine-tune on that to close the gap between the public data and our real deployment conditions.